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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/037,592	05/18/2023	Jonathan Michael BLACKBURN	16742S-000003-US-NP	5069

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EXAMINER
COUNTS, GARY W

ART UNIT	PAPER NUMBER
1678	

NOTIFICATION DATE	DELIVERY MODE
04/23/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

troymailroom@hdp.com

Office Action Summary	Application No. 18/037,592	Applicant(s) BLACKBURN et al.	
	Examiner GARY W COUNTS	Art Unit 1678	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 04/03/26.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-9,11-12 and 20-26 is/are pending in the application.

5a) Of the above claim(s) 20-26 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-9 and 11-12 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☑ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 05/18/23 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☑ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 08/09/23, 03/21/25, 04/22/25.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Election/Restrictions

Applicant's election of Group I, Claims 1-9 and 11-12 in the reply filed on 04/03/26 is acknowledged. Because applicant did not distinctly and specifically point out the supposed errors in the restriction requirement, the election has been treated as an election without traverse (MPEP § 818.01(a)). Currently, claims 1-9, 11-12 and 20-26 are pending. Claims 20-26 are withdrawn as being directed to non-elected inventions. Accordingly, claims 1-9 and 11-12 are under examination.

Information Disclosure Statement

The listing of references in the specification is not a proper information disclosure statement. 37 CFR 1.98(b) requires a list of all patents, publications, or other information submitted for consideration by the Office, and MPEP § 609.04(a) states, "the list may not be incorporated into the specification but must be submitted in a separate paper." Therefore, unless the references have been cited by the examiner on form PTO-892, they have not been considered.

Specification

The specification has not been checked to the extent necessary to determine the presence of all possible minor errors. Applicant's cooperation is requested in correcting any errors of which applicant may become aware in the specification.

Claim Objections

Claim 1 is objected to because of the following informalities:

Claim 1, line 1 the recitation "serum/plasma" should be --serum or plasma--.

Claim 1, step (ii) the recitation "characterized in that said" is objected to because the recitation "characterized in that it" is not a commonly recognized transitional phrase used in U.S. practice, it is suggested to change the recitation to --wherein said--.

Appropriate correction is required.

Claim Rejections - 35 USC § 101

35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

Claims 1-9 and 11-12 are rejected under 35 U.S.C. 101 because the claimed invention is directed to abstract ideas and/or to laws of nature/natural phenomena without significantly more.

The U.S. Patent and Trademark Office recently revised the MPEP with regard to § 101 (see the MPEP at 2106). Regarding the MPEP at 2106, in determining what concept the claim is "directed to," we first look to whether the claim recites:

(1) any judicial exceptions, including certain groupings of abstract ideas (*i.e.*, mathematical concepts, certain methods of organizing human activity such as a fundamental economic practice, or mental processes); and

(2) additional elements that integrate the judicial exception into a practical application (see MPEP § 2106.05(a)-(c), (e)-(h)).

Only if a claim (1) recites a judicial exception and (2) does not integrate that exception into a practical application, do we then look to whether the claim contains an "'inventive concept' sufficient to 'transform'" the claimed judicial exception into a patent-eligible application of the judicial exception. *Alice*, 573 U.S. at 221 (quoting *Mayo*, 566 U.S. at 82). In so doing, we thus consider whether the claim:

(3) adds a specific limitation beyond the judicial exception that is not "well-understood, routine, conventional" in the field (see MPEP § 2106.05(d)); or

(4) simply appends well-understood, routine, conventional activities previously known to the industry, specified at a high level of generality, to the judicial exception.
See MPEP 2106.

ELIGIBILITY STEP 2A: WHETHER A CLAIM IS DIRECTED TO A JUDICIAL EXCEPTION

Step 2A, Prong 1

The claims are directed to a naturally occurring correlation between autoantibodies in a serum/plasma sample of a rheumatoid arthritis patient prior to treatment with adalimumab and the patient having a good response or a poor response.

Step 2A, Prong 2

The additional elements of testing the sample for autoantibodies to SSB, TROVE2 and ZHX2 does not apply, rely on, or use the judicial exception in a manner that imposes a meaningful limit on the judicial exception.

Also, with respect to the recitation “determining whether the patient will develop a good response or a bad poor response to treatment with adalimumab, based on the detection of said autoantibody biomarkers; characterized in that said autoantibody biomarkers comprise autoantibodies to antigens SSB, TROVE2 and ZHX2, wherein ZHX2 is associated with the good response, and SSB and TROVE2 are associated with the poor response”. The “determining” statement at best articulates the judicial exception, amounting only to a general instruction to apply or use the judicial exception. This could read on mental activity being performed solely in a practitioner’ head, e.g. A mental appreciation of the recited autoantibodies being correlated a good response or a poor response to adalimumab treatment in the subject. No active method steps are invoked or clearly required; the “determining” statement does not include any activity that would constitute a practical application, i.e. steps that apply, rely on or use the natural principle in a manner such that the claims amount to significantly more that the natural principal itself.

ELIGIBILITY STEP 2B: WHETHER THE ADDITIONAL ELEMENTS CONTRIBUTE AN "INVENTIVE CONCEPT"

Further, the additional elements of the claims are recited with a high level of generality and do not apply, rely on, or use the judicial exception in a manner that imposes a meaningful limit on the judicial exception. (the active method

steps/limitations recited in addition to the judicial exceptions themselves) and do not add significantly more to the judicial exception(s).

As shown by ImmuneTM (submitted in the IDS filed 08/29/23) it is well known, routine and conventional in the art to test a sample for autoantibodies to SSB, TROVE2 and ZHX2 and also additional autoantibody biomarkers (e.g. pages 1, 42, 46, 48).

It does not appear to be the case that the active steps recited, which are performed in order to gather the data or perform the assay, are steps recited or performed in an unconventional or non-routine way, such to provide an inventive concept under step 2B.

The claimed limitations as currently presented fail to recite limitations that add a feature that is more than well understood, conventional or routine in the field of diagnostics and biochemical assay methodologies.

For all of these reasons, the claims fail to include additional elements that are sufficient to either integrate the judicial exception(s) into practical application(s) thereof, or amount to significantly more than the judicial exception(s).

Allowable Subject Matter

Claims 1-9 and 11-12 would be allowable if rewritten or amended to overcome the rejection(s) under 35 U.S.C. 101, set forth in this Office action. The prior art of record does not teach nor fairly suggest testing the serum/plasma sample for the presence of autoantibodies to antigens SSB, TROVE2 and ZHX2 and correlating the presence to a good or poor response to adalimumab in the patient as currently recited.

Conclusion

No claims are allowed.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to GARY W COUNTS whose telephone number is (571)272-0817. The examiner can normally be reached M-F 7:00-4:00.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Gregory Emch can be reached at 571-272-8149. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/GARY COUNTS/
Primary Examiner, Art Unit 1678